



**CURIOSITY
CREATES**

**A BETTER
WORLD**

TOPIC 5 – PHYSICAL PROPERTIES

QEng Prep - Chemistry



FACULTY OF
ENGINEERING AND
APPLIED SCIENCE

KEY CONCEPTS

The key concepts covered in this lecture include:

- Explain how the physical properties of a solid or liquid depend on the particles present;
 - Explain how the physical properties of a solid or liquid depend the types of intermolecular and intramolecular forces (e.g., covalent bonding, ionic bonding, Van der Waals forces, hydrogen bonding, metallic bonding) present; and
 - Understand and explain the different physical properties of various solid structures, such as metallic solids, ionic solids, molecular solids, covalent network solids.
-

FORCES AND BONDING



INTRODUCTION

The physical properties of a substance depend on its structure and type of bonding present, and bonding determines the type of structure.

Basic Theory

- The noble gases (He, Ne, Ar, Kr, Xe and Rn) of Group 18 are all relatively (or totally) inert
 - Their electronic structure appears to confer stability, as they have just filled their valence shell of electrons
 - Atoms without the electron structure of a noble gas try to get one
 - There are various ways for atoms to achieve stability, the method depends on an element's position in the periodic table
 - There are two types of bonds:
 1. Chemical bonds:
 - These bonds are strong;
 - Examples include ionic bonds, covalent bonds and metallic bonds
 2. Physical bonds:
 - These bonds are weak;
 - Examples include Van der Waals forces (weakest) and hydrogen bonds (strongest).
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IONIC BONDING

Definition

- Compounds composed of ions are called ionic compounds (or salts), and their constituent ions are held together by ionic bonds (i.e. electrostatic forces of attraction between oppositely charged cations and anions)

Formation

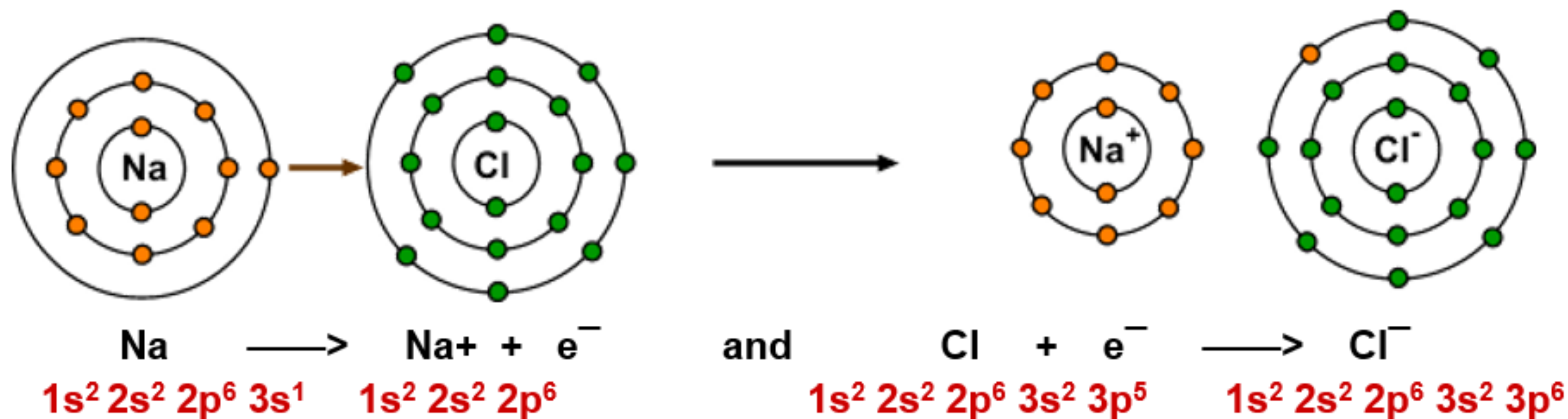
- An ionic bond is formed when a metal donates its valence electrons to a nonmetal so that both elements are able to gain the nearest noble gas configuration
- Many metallic elements have relatively low ionization potentials and lose electrons easily. These elements lie to the left in a period or near the bottom of a group on the Periodic Table. Nonmetal atoms have relatively high electron affinities and thus readily gain electrons lost by metal atoms, thereby filling their valence shells. Nonmetallic elements are found in the upper-right corner of the Periodic Table.
- The attractive forces between ions are isotropic (i.e., the same in all directions) meaning that any particular ion is equally attracted to all of the nearby ions of opposite charge. This results in the ions arranging themselves into a **tightly bound, three-dimensional lattice structure**.

- [illegible]

IONIC BONDING

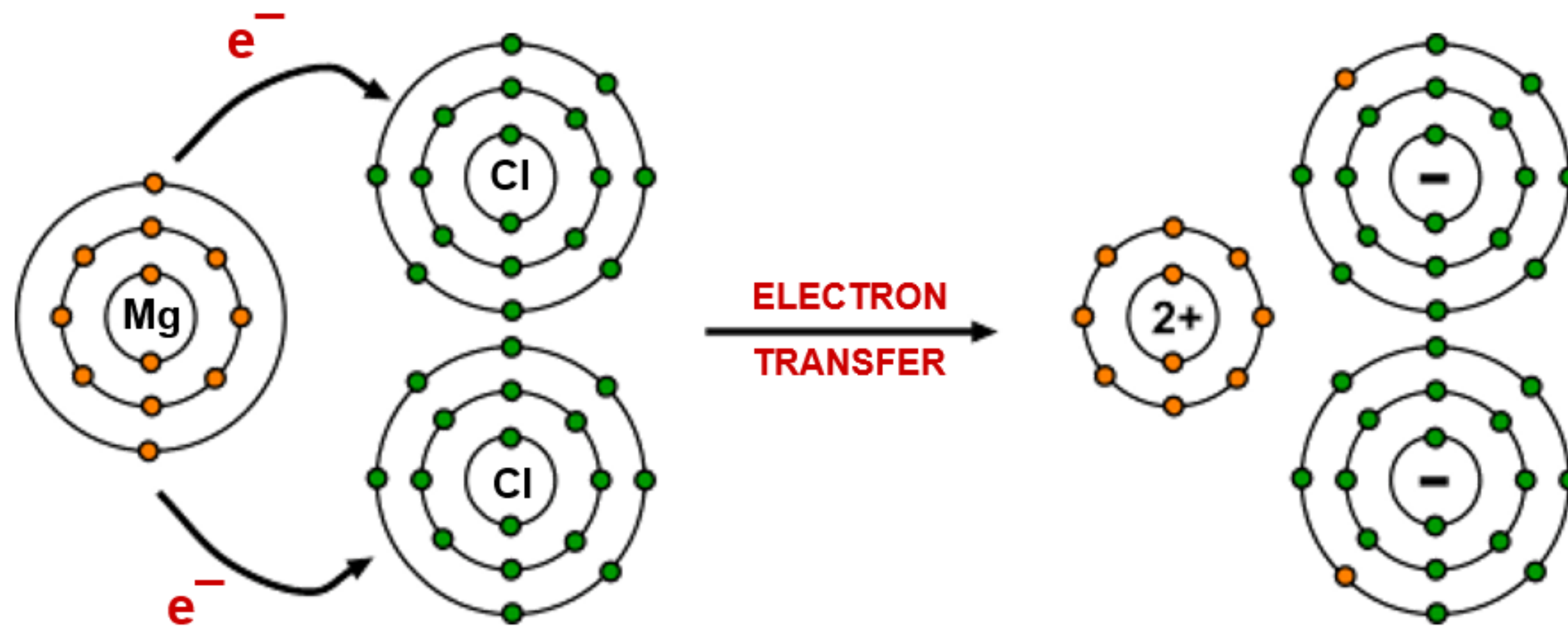
As mentioned earlier, ionic bonds tend to be formed between elements whose atoms need to “lose” electrons to gain the nearest noble gas electronic configuration and those which need to gain electrons. The electrons are transferred from one atom to the other.

For example, for sodium chloride, NaCl, An electron is transferred from the 3s orbital of sodium to the 3p orbital of chlorine; both species end up with the electronic configuration of the nearest noble gas the resulting ions are held together in a crystal lattice by electrostatic attraction.



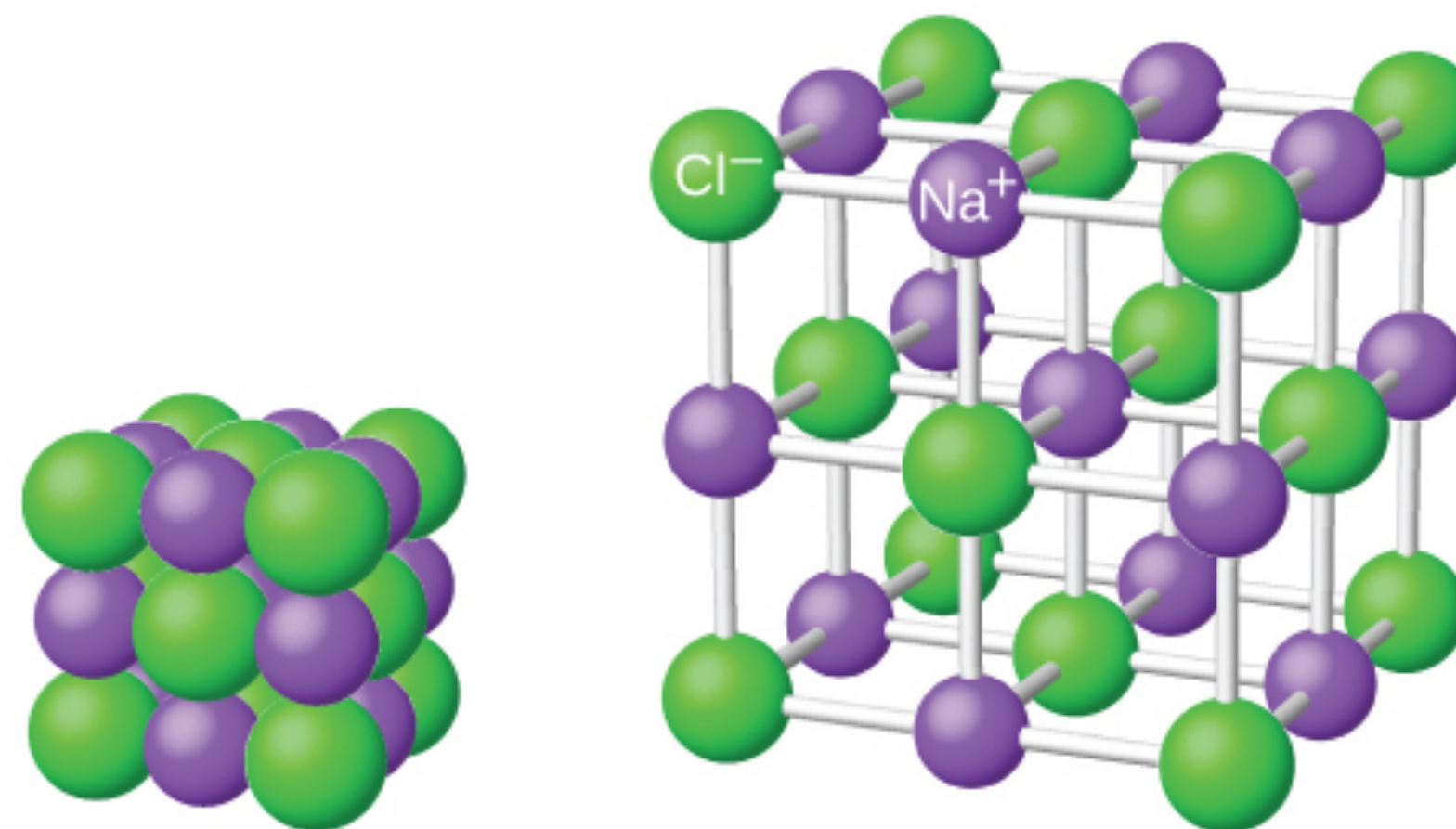
IONIC BONDING

Another example would be the formation of magnesium chloride, MgCl_2 :



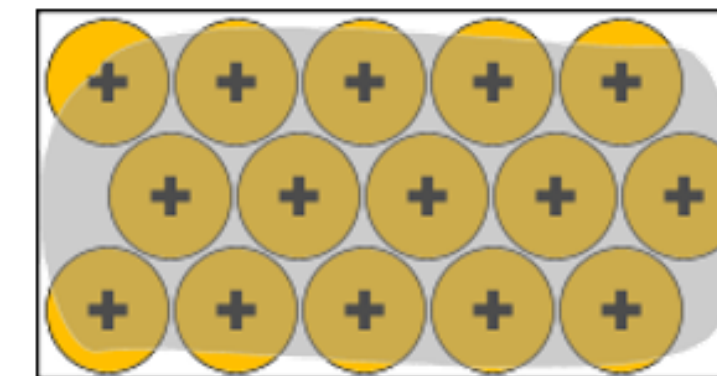
IONIC BONDING

- It is important to note that the **formula for an ionic compound does not represent the physical arrangement of its ions**. It is incorrect to refer to a sodium chloride (NaCl) “molecule” because there is not a single ionic bond, per se, between any specific pair of sodium and chloride ions.
- For example, sodium chloride consists of a regular arrangement of equal numbers of Na⁺ cations and Cl⁻ anions. The atoms in sodium chloride are arranged to maximize opposite charges interacting (on the left). The smaller spheres represent sodium ions, the larger ones represent chloride ions. In the expanded view (on the right), the geometry can be seen more clearly. Note that each ion is “bonded” to all of the surrounding ions (six in this case).



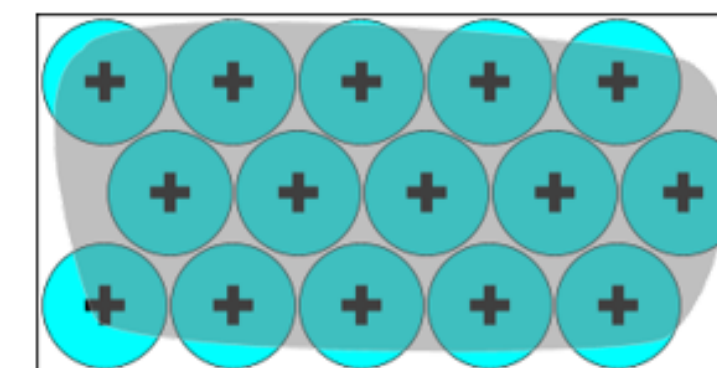
METALLIC BONDING

- Involves a lattice of positive ions surrounded by delocalized electrons. Metal atoms achieve stability by “off-loading” electrons to attain the electronic structure of the nearest noble gas. These electrons join up to form a mobile cloud which prevents the newly-formed positive ions from flying apart due to repulsion between similar charges.
- Atoms arrange in regular close-packed 3-dimensional crystal lattices. The outer shell electrons of each atom leave to join a mobile “cloud” or “sea” of electrons which can roam throughout the metal. The electron cloud binds the newly-formed positive ions together.
- Metallic bond strength depends on the number of outer electrons donated to the cloud, and the size of the metal atom or ion. For example, the bond strengths of sodium, potassium and magnesium are compared to the right:



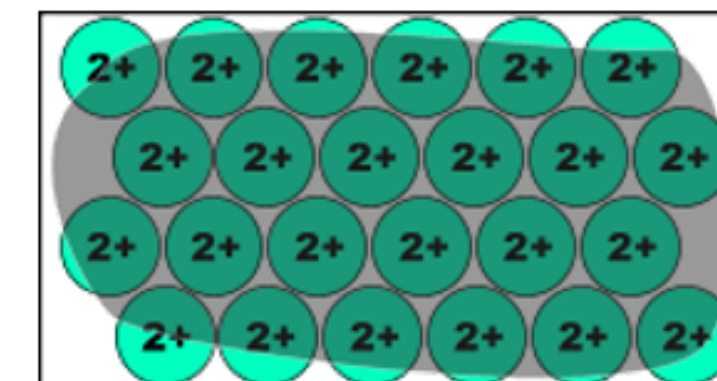
Na

The strength of the metallic bonding in sodium is relatively weak because each atom donates one electron to the cloud.



K

The metallic bonding in potassium is weaker than in sodium because the resulting ion is larger and the electron cloud has a bigger volume to cover so is less effective at holding the ions together.



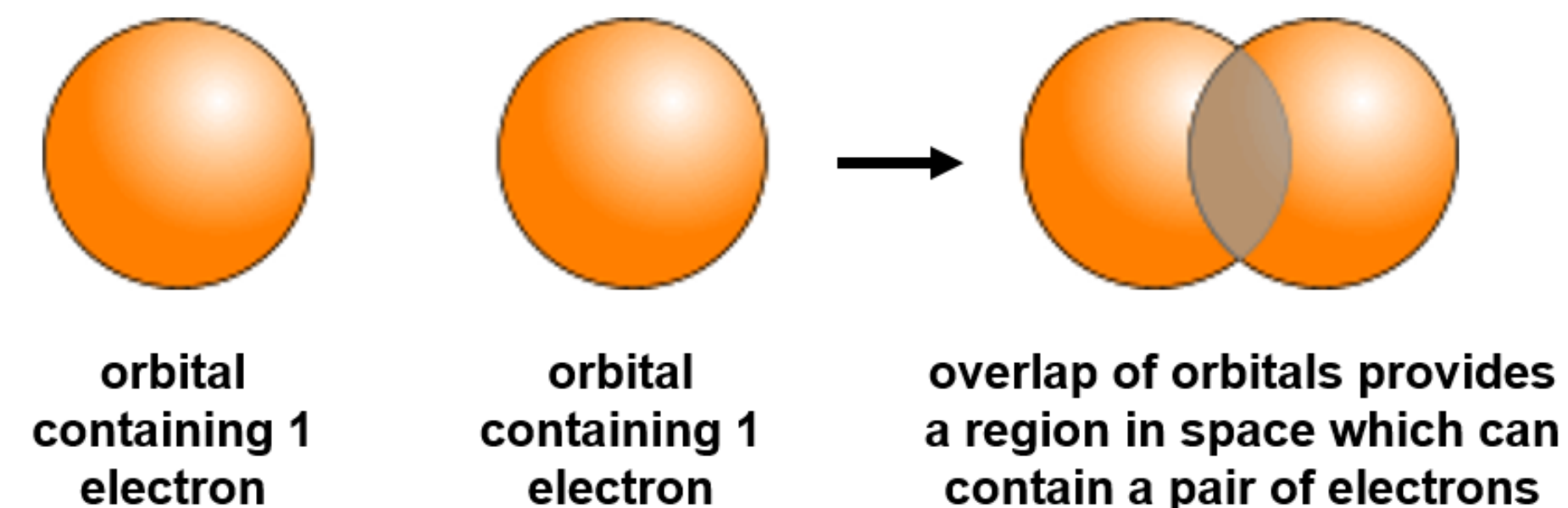
Mg

The metallic bonding in magnesium is stronger than in sodium because each atom has donated two electrons to the cloud. The greater the electron density holds the ions together more strongly.

COVALENT BONDING

Definition

- A type of bonding which results from the mutual attraction of atoms for a “shared” pair of electrons. Covalent bonds are formed between two atoms when both have similar tendencies to attract electrons to themselves (i.e. when both atoms have identical or fairly similar ionization energies and electron affinities).
- Typically atoms share electrons to get the nearest noble gas electron configuration. For example, two hydrogen atoms bond covalently to form an H_2 molecule; each hydrogen atom in the H_2 molecule has two electrons stabilizing it, giving each atom the same number of valence electrons as the noble gas He:



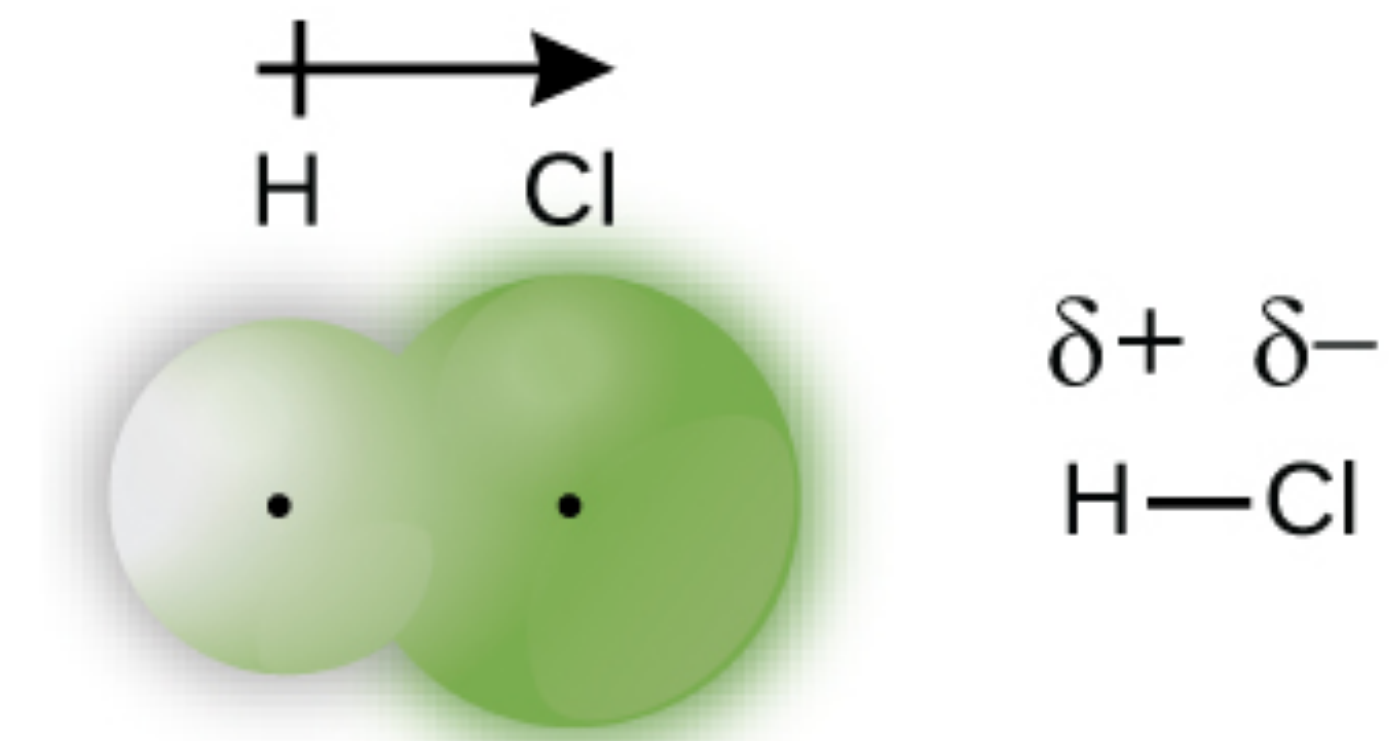
COVALENT BONDING

Properties

- Compounds that contain covalent bonds exhibit different physical properties than ionic compounds.
 - Because the attraction between molecules, which are electrically neutral, is weaker than that between electrically charged ions, covalent compounds generally have much lower melting and boiling points than ionic compounds.
 - Many covalent compounds are liquids or gases at room temperature, and, in their solid states, they are typically much softer than ionic solids.
 - Most covalent compounds are insoluble in water.
 - Since they are electrically neutral, they are poor conductors of electricity in any state.

PURE VS. POLAR COVALENT BONDS

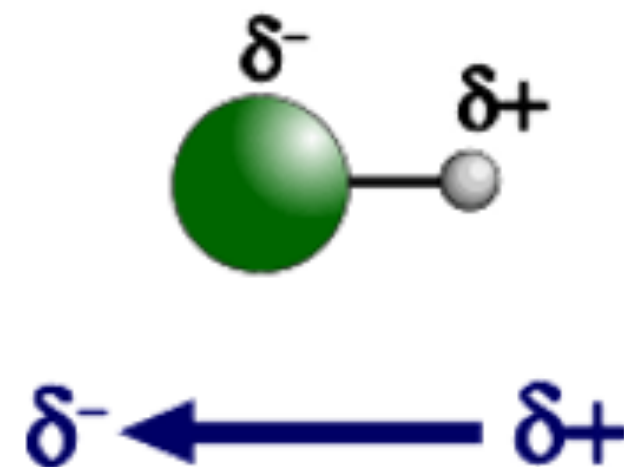
- **Pure covalent bonds** are formed when the electrons in a bond are shared equally, and thus have an equal probability of being near each nucleus. This occurs if the atoms forming the bond are identical (i.e. one of the seven diatomic molecules).
- When the atoms linked by a covalent bond are different, the bonding electrons are shared, but no longer equally. Instead, the bonding electrons are more attracted to one atom than the other, giving rise to a shift of electron density toward that atom. This unequal distribution of electrons is known as a **polar covalent bond**, characterized by a partial positive charge on the atom that attracts the electrons more strongly and vice versa.
- We designate the positive and negative atoms in a polar covalent bond using a lowercase Greek letter “delta,” δ , with a plus sign or minus sign to indicate whether the atom has a partial positive charge ($\delta+$) or a partial negative charge ($\delta-$).



POLAR MOLECULES

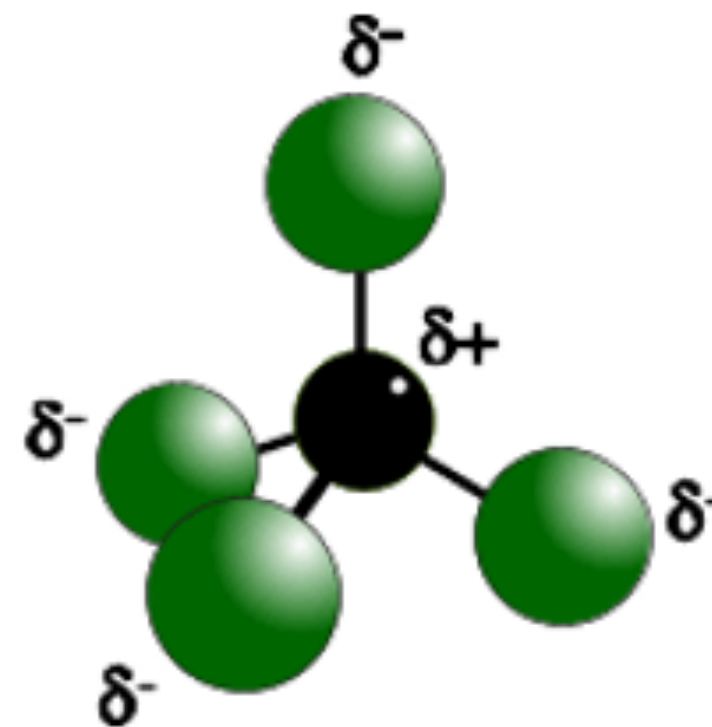
- Not all molecules containing polar bonds are polar overall.
- If bond dipoles 'cancel each other' the molecule isn't polar, but if there is a 'net dipole' the molecule will be polar:

HYDROGEN CHLORIDE



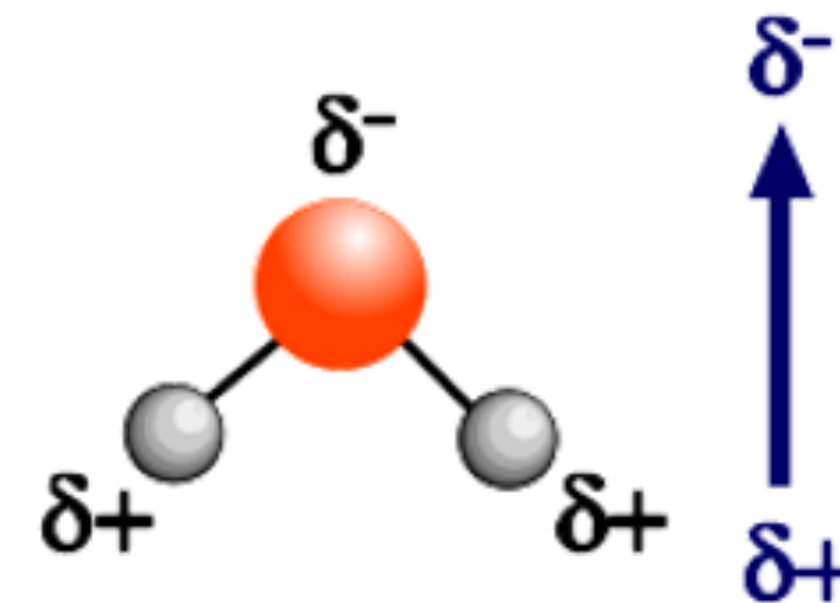
NET DIPOLE - POLAR

TETRACHLOROMETHANE



NON-POLAR

WATER

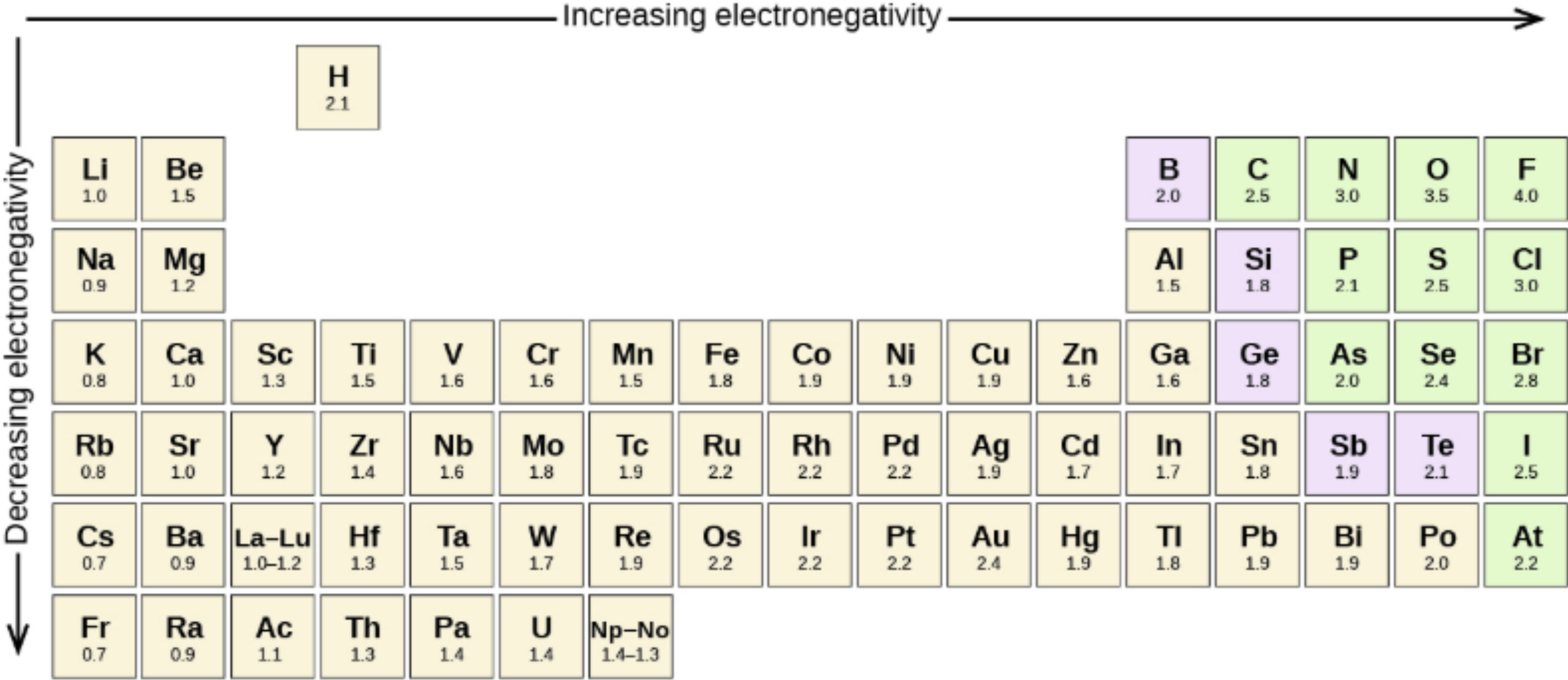


NET DIPOLE - POLAR

ELECTRONEGATIVITY AND COVALENT BONDS

- Whether a bond is nonpolar or polar covalent is determined by a property of the bonding atoms called **electronegativity**.
- Electronegativity is a measure of the tendency of an atom to attract electrons (or electron density) towards itself. It determines how the shared electrons are distributed between the two atoms in a bond.
- The figure on the next slide shows the electronegativity values of the elements as proposed by Linus Pauling.
- In general, electronegativity:
 - Increases from left to right across a period in the periodic table; and
 - Decreases down a group.
- The nonmetals, which lie in the upper right, tend to have the highest electronegativities, with fluorine being the most electronegative element of all (EN = 4.0).
- Metals tend to be less electronegative elements, and the Group 1 metals have the lowest electronegativities.
- Note that noble gases are excluded from this figure because these atoms usually do not share electrons with other atoms since they have a full valence shell.

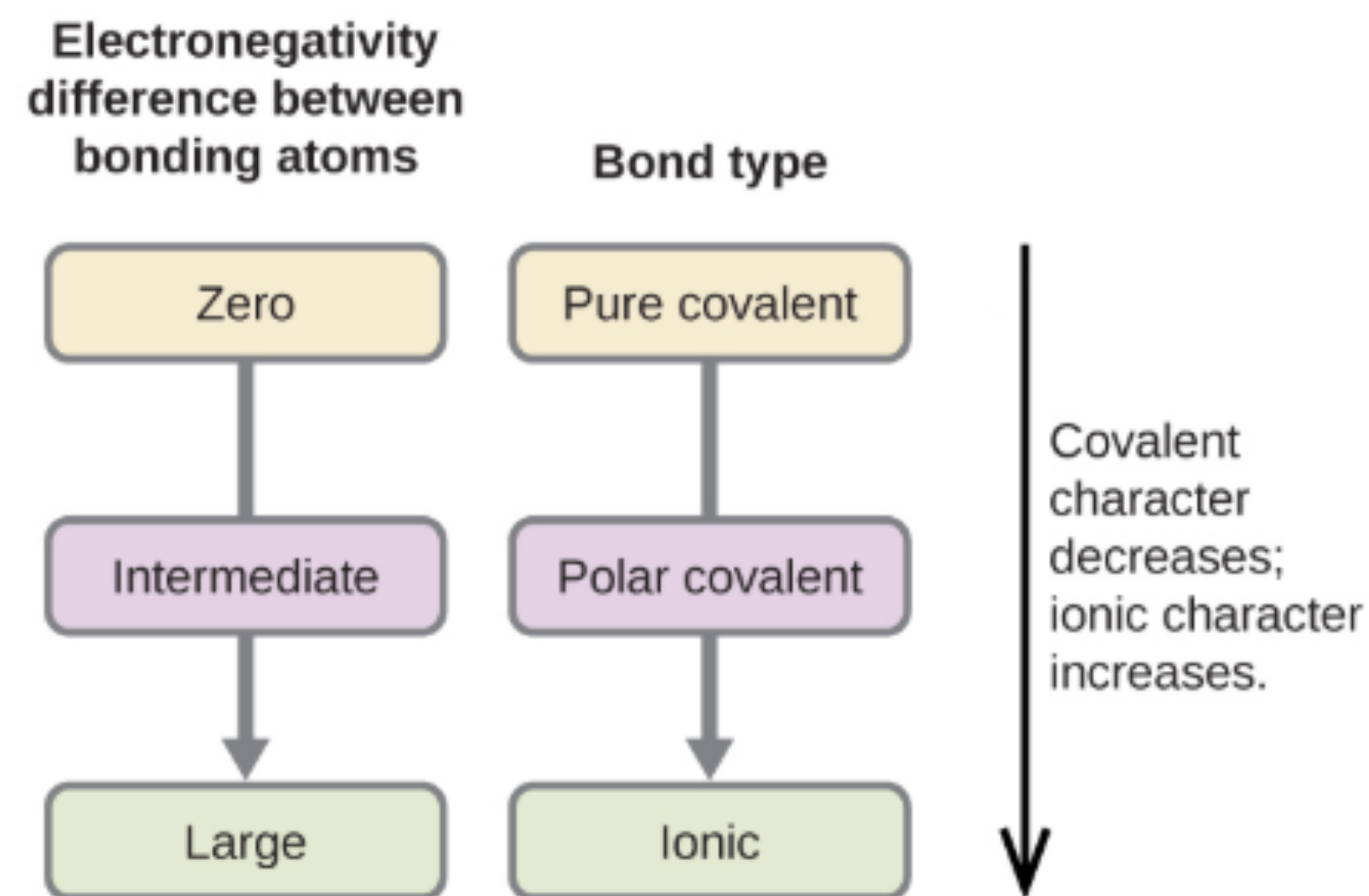
ELECTRONEGATIVITY AND COVALENT BONDS



ELECTRONEGATIVITY AND BOND TYPE

- The more strongly an atom attracts the electrons in its bonds, the larger its electronegativity. Electrons in a polar covalent bond are shifted toward the more electronegative atom; as such, the more electronegative atom is the one with the partial negative charge.
- The absolute value of the difference in electronegativity (ΔEN) of two bonded atoms provides a rough measure of the polarity to be expected in the bond and, as such, the bond type.

- The greater the difference in electronegativity, the more polarized the electron distribution will be resulting in larger partial charges of the atoms:
- This means:
 - ΔEN small or zero $\rightarrow$ the bond is covalent and nonpolar
 - ΔEN large $\rightarrow$ the bond is polar covalent or ionic.



Bond Type	Electronegativity Difference
pure covalent	< 0.4
polar covalent	between 0.4 and 1.8
ionic	> 1.8

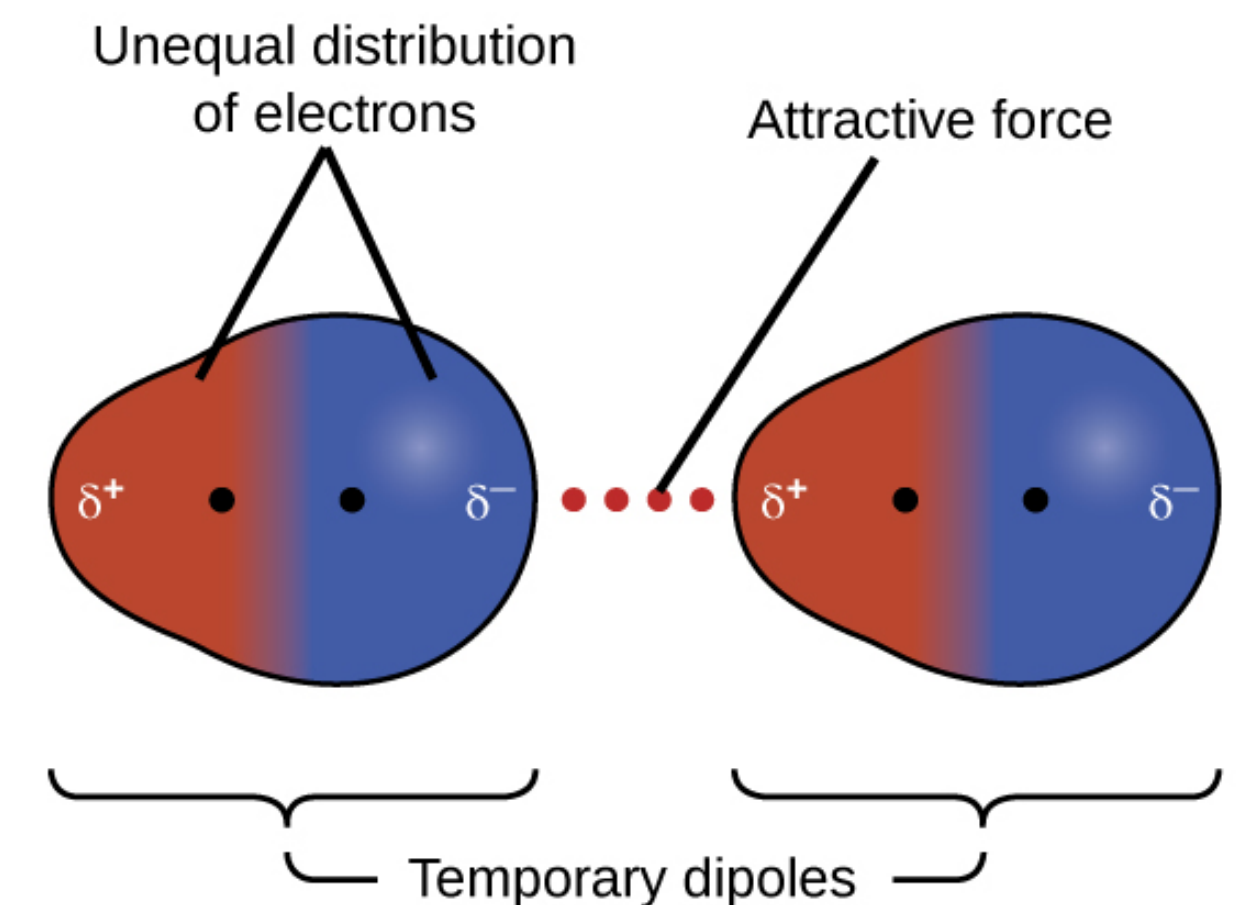
VAN DER WAALS FORCES

Definition

- Van der Waals forces is a general term used to define the attraction of intermolecular forces between molecules. Although the bonding *within* molecules is strong, that *between* molecules is weak (molecules and monatomic noble gases are subject to weak attractive forces)
- There are two kinds of Van der Waals forces:
 1. Weak London Dispersion Forces
 2. Stronger dipole-dipole forces.

London Dispersion Forces: temporary attractive forces that result when the electron density of a molecule results in a dipole. Because the position of electrons in an atom is constantly changing, there is a possibility that at any given time one side will have more electrons than the other. This creates a dipole that then induces dipoles in adjacent molecules.

Dipole-Dipole Forces: similar to London Dispersion forces, but they occur in molecules that are permanently polar versus momentarily polar. A polar molecule attracts the positive end of another polar molecule with the negative end of its dipole.



VAN DER WAALS FORCES

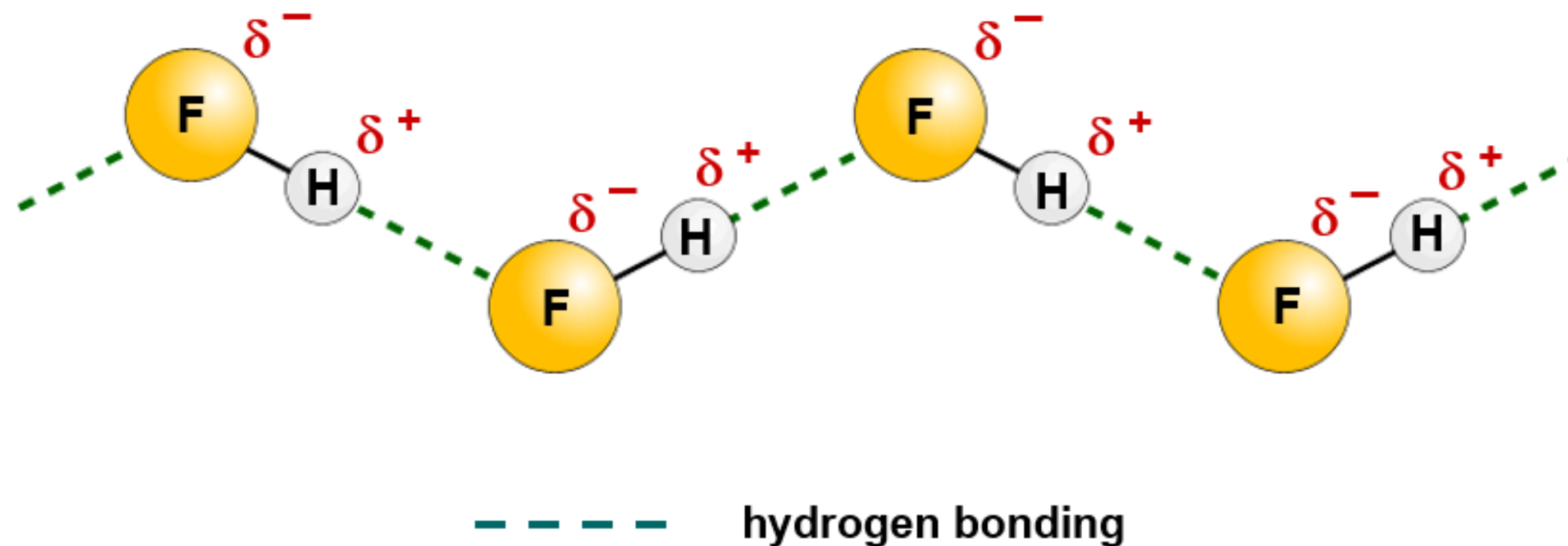
Trends and Properties

- The more electrons a molecule contains, the higher its ability to become polar. Polarizability increases in the periodic table from the top of a group to the bottom and from right to left within periods. This is because the higher the molecular mass, the more electrons an atom has. With more electrons, the outer electrons are easily displaced because the inner electrons shield the nucleus' positive charge from the outer electrons which would normally keep them close to the nucleus.
- When the molecules become polar, the melting and boiling points are raised because it takes more heat and energy to break these bonds. Therefore, the greater the mass, the more electrons present, and the more electrons present, the higher the melting and boiling points of these substances.

Melting and Boiling Points of the Halogens				
Halogen	Molar Mass	Atomic Radius	Melting Point	Boiling Point
fluorine, F ₂	38 g/mol	72 pm	53 K	85 K
chlorine, Cl ₂	71 g/mol	99 pm	172 K	238 K
bromine, Br ₂	160 g/mol	114 pm	266 K	332 K
iodine, I ₂	254 g/mol	133 pm	387 K	457 K
astatine, At ₂	420 g/mol	150 pm	575 K	610 K

HYDROGEN BONDING

- Hydrogen bonding is an extension of dipole-dipole interaction that occurs between H and the three most electronegative elements: F, O and N. These bonds are extremely polar.
- Because of the small sizes of H, F, N and O, the partial charges are concentrated in a small volume which leads to a high charge density and makes the intermolecular attractions greater. This leads to even higher boiling points.
- For example, hydrogen fluoride (HF) has a much higher boiling point than one would expect for a molecule with a relative molecular mass of 20. Fluorine has the highest electronegativity of all atoms, and is a small atom so the bonding with hydrogen is extremely polar:



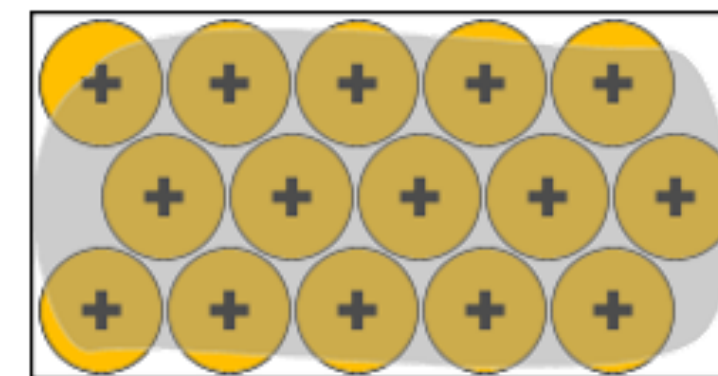
SOLID STRUCTURES



METALLIC SOLIDS

- A metallic solid is a solid with the **characteristic properties of a metal**:
 - Shiny and silvery in color;
 - A good conductor of heat and electricity; and
 - Can also be hammered into sheets (malleable) and pulled into wires/rods (ductile).
- A metallic solid exhibits metallic bonding, a type of intermolecular interaction caused by the **sharing of the s valence electrons** by all atoms.
- For a substance to conduct electricity it must have mobile ions or electrons. Because the electron cloud is mobile, electrons are free to move throughout its structure, allowing it to conduct electricity (electrons attracted to the positive end are replaced by those entering from the negative end).

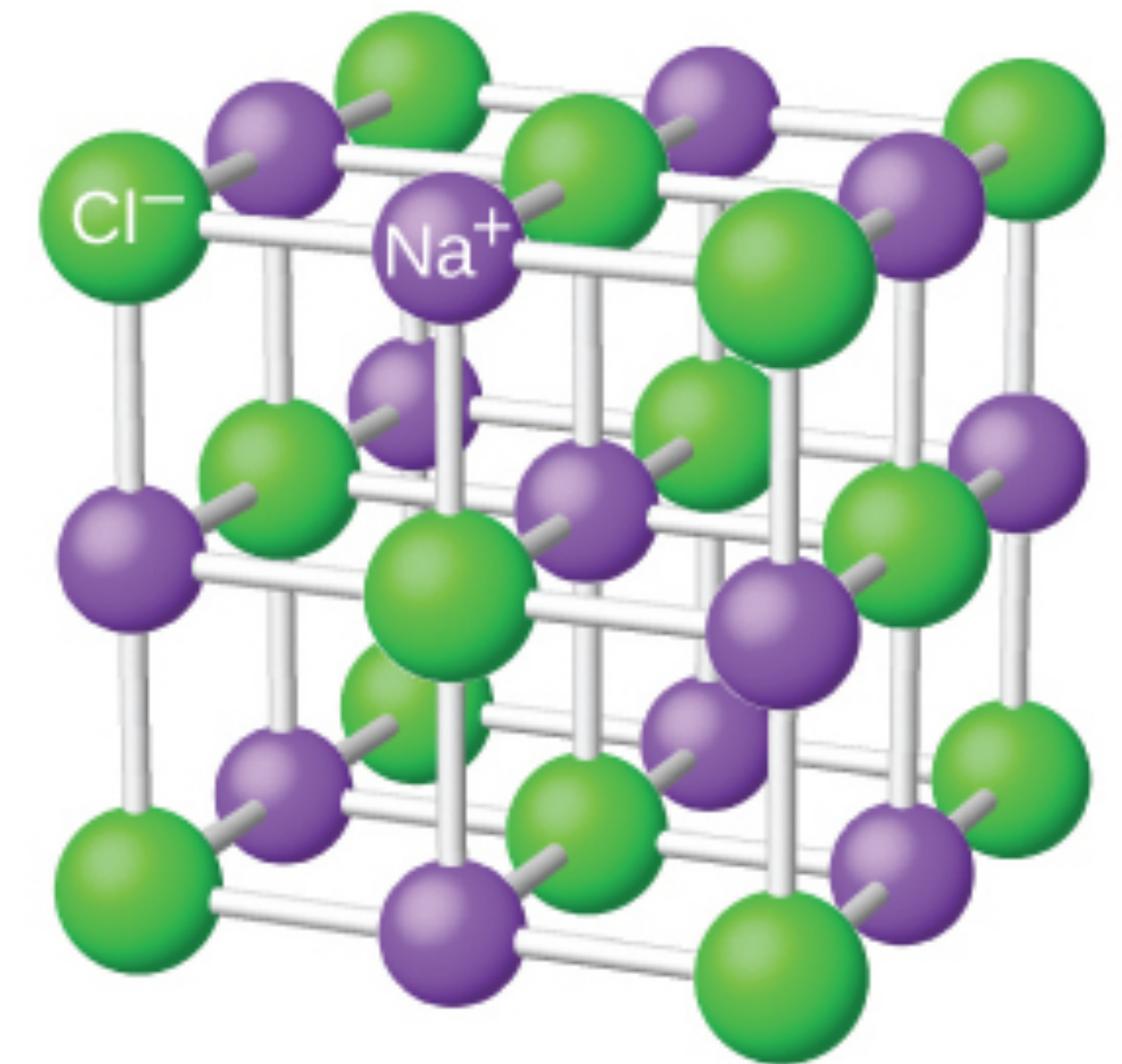
As the metal is hammered into another shape the delocalised electron cloud continues to bind the “ions” together.



Some metals, such as gold, can be hammered into sheets thin enough to be translucent.

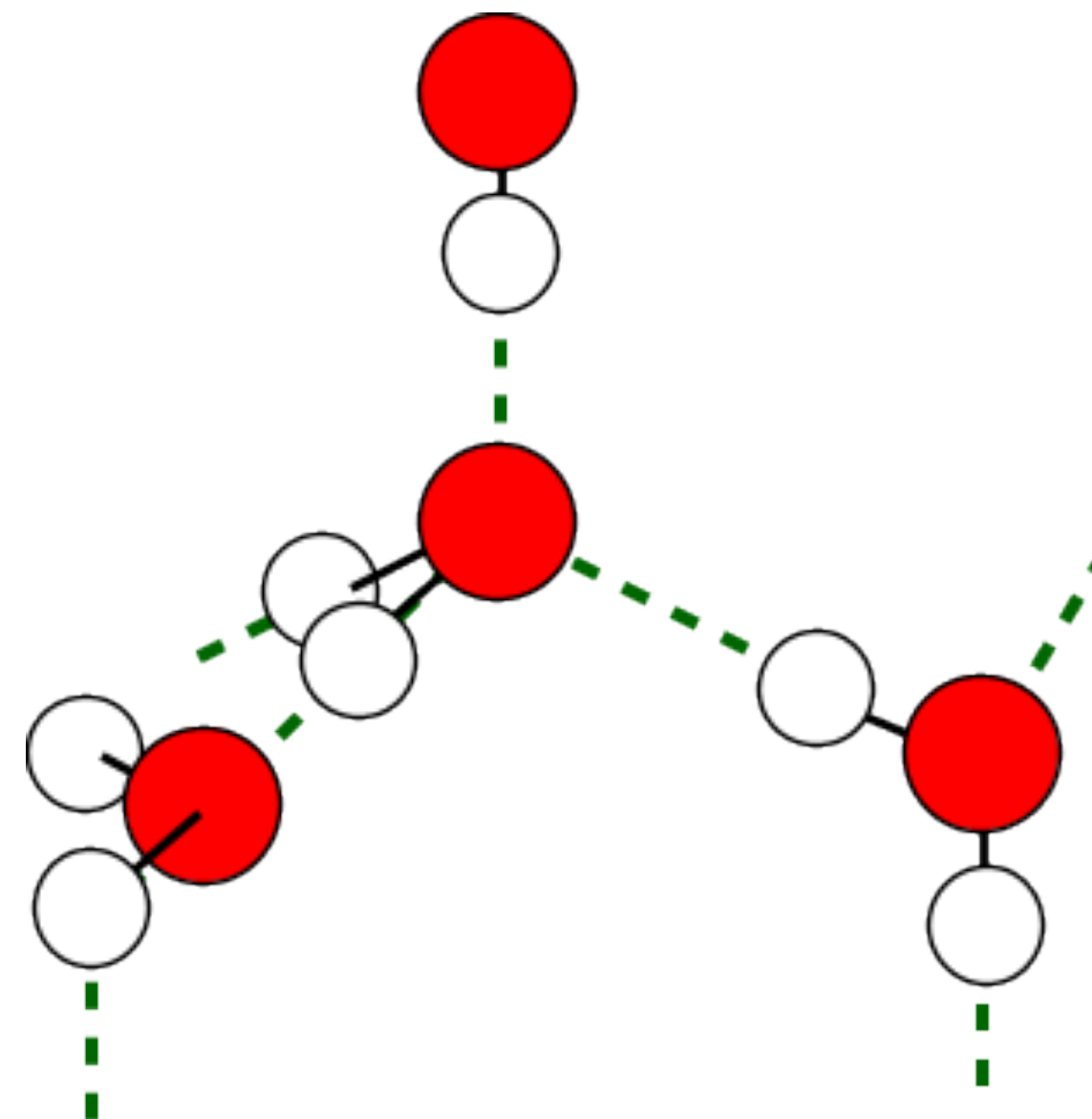
IONIC SOLIDS

- Compounds composed of ions are called ionic solids, and their constituent ions are held together by electrostatic forces of attraction between oppositely charged cations and anions.
- Ionic solids exhibit a crystalline (i.e. lattice) structure and tend to be rigid and brittle; they also tend to have high melting and boiling points, which suggests that ionic bonds are very strong.
- Ionic solids are also poor conductors of electricity for the same reason - the strength of ionic bonds prevents ions from moving freely in the solid state.
- Most ionic solids dissolve readily in water. Once dissolved or melted, ionic compounds are excellent conductors of electricity and heat because the ions can move about freely.



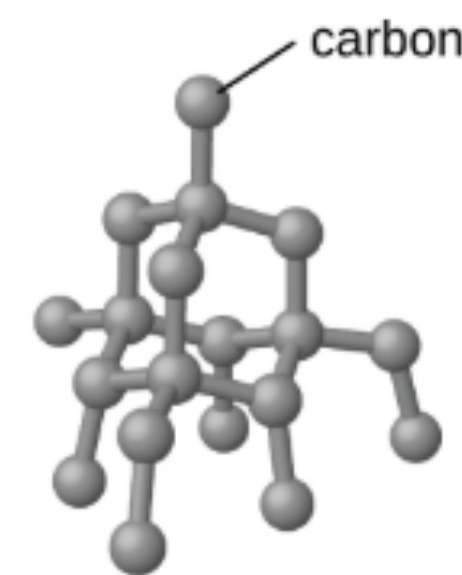
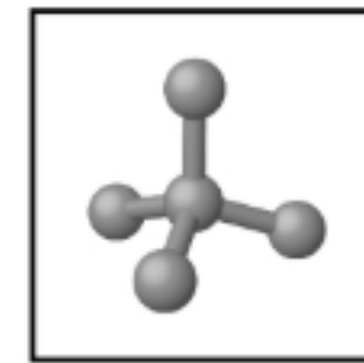
MOLECULAR SOLIDS

- A molecular solid is a crystalline solid whose components are covalently bonded molecules.
- Many molecular substances form solids where the molecules line up in a similar way to ionic solids; however, they are made up of molecules instead of ions. Because the intermolecular forces between molecules are typically less strong than in ionic solids, molecular solids typically melt at lower temperatures and are softer than ionic solids.
- For example, ice is a molecular solid. In the solid state, the molecules line up in a regular pattern:

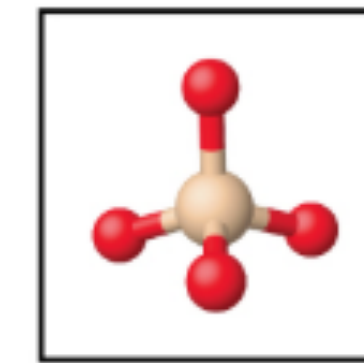


COVALENT NETWORK SOLIDS

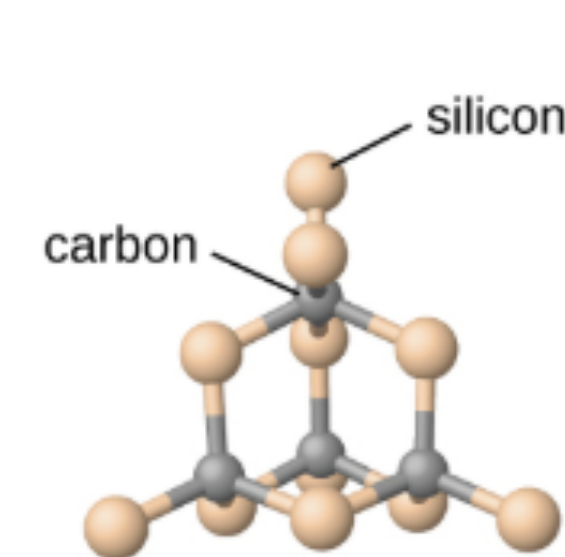
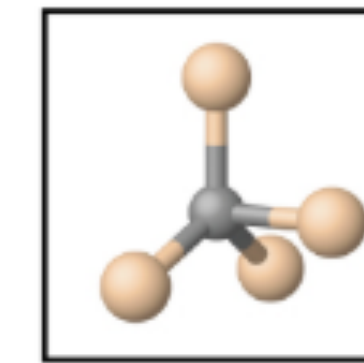
- Covalent network solids include crystals of diamond, silicon, some other nonmetals, and some covalent compounds such as silicon dioxide (sand). Many minerals have networks of covalent bonds.
- The atoms in these solids are held together by a network of covalent bonds, as shown in the figure below.
- To break or to melt a covalent network solid, covalent bonds must be broken. Because covalent bonds are relatively strong, covalent network solids are typically characterized by hardness, strength, and high melting points. For example, diamond is one of the hardest substances known and melts above 3500°C.



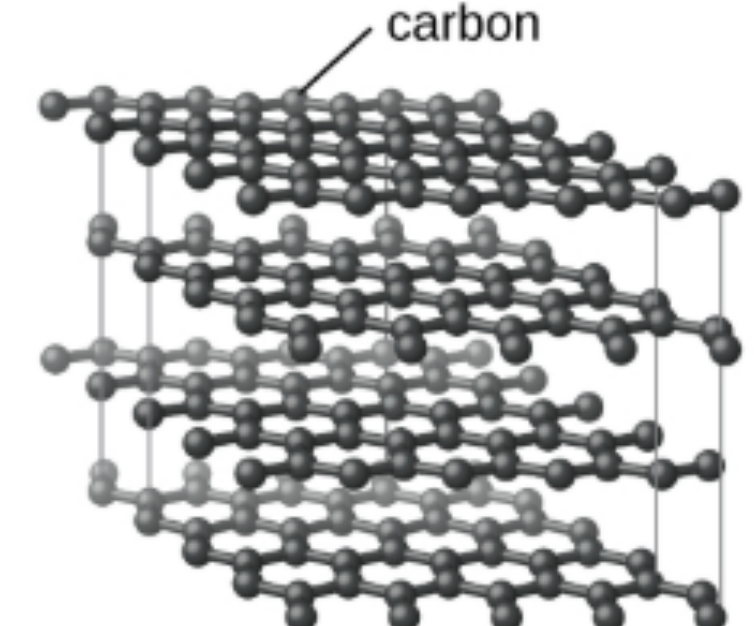
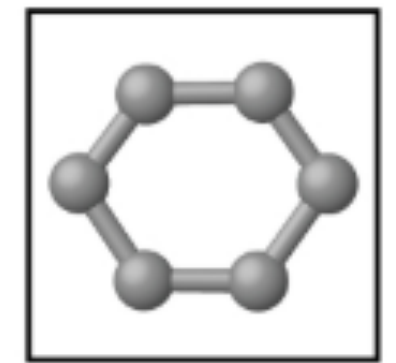
diamond



silicon dioxide



silicon carbide



graphite

PHYSICAL PROPERTIES



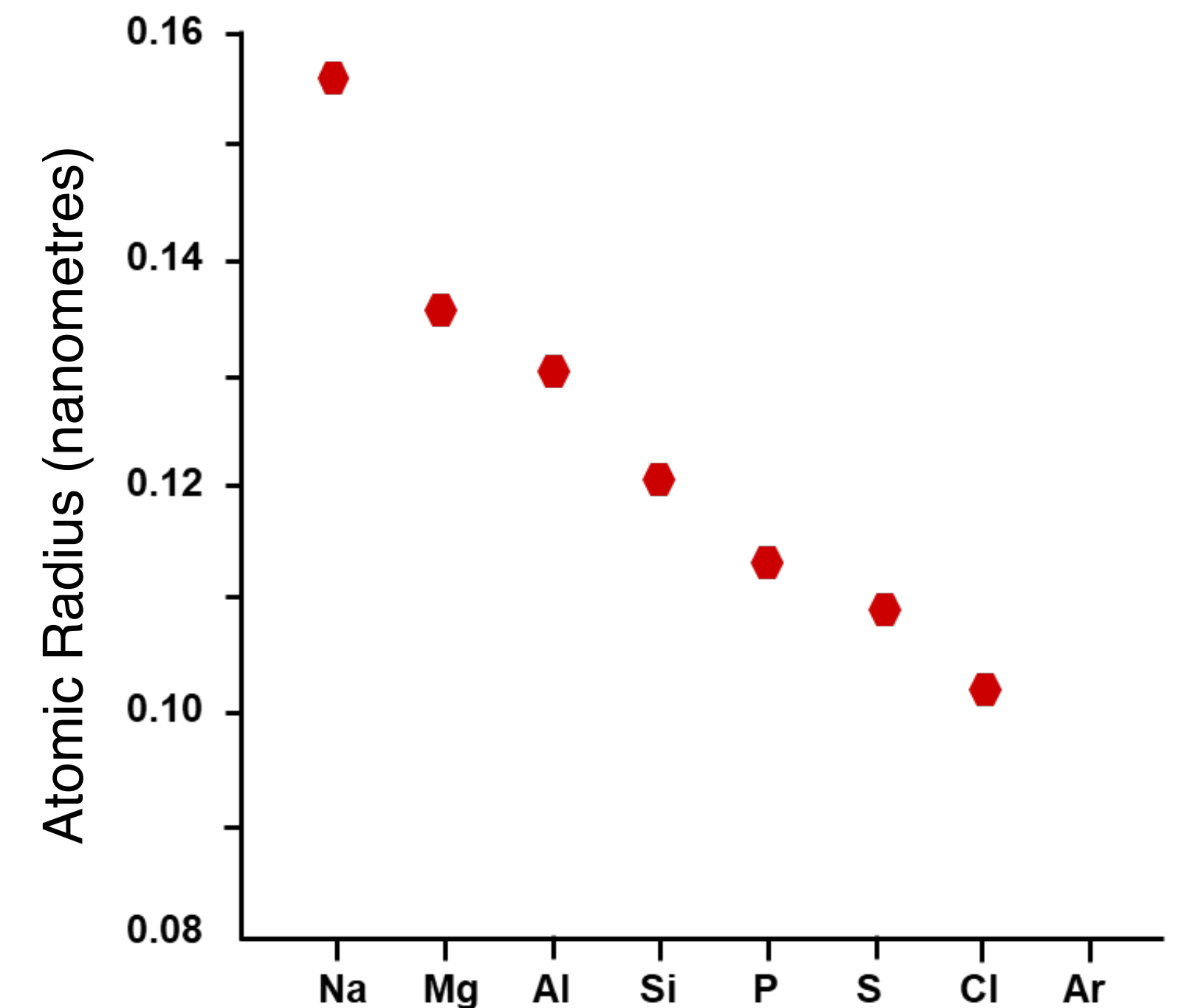
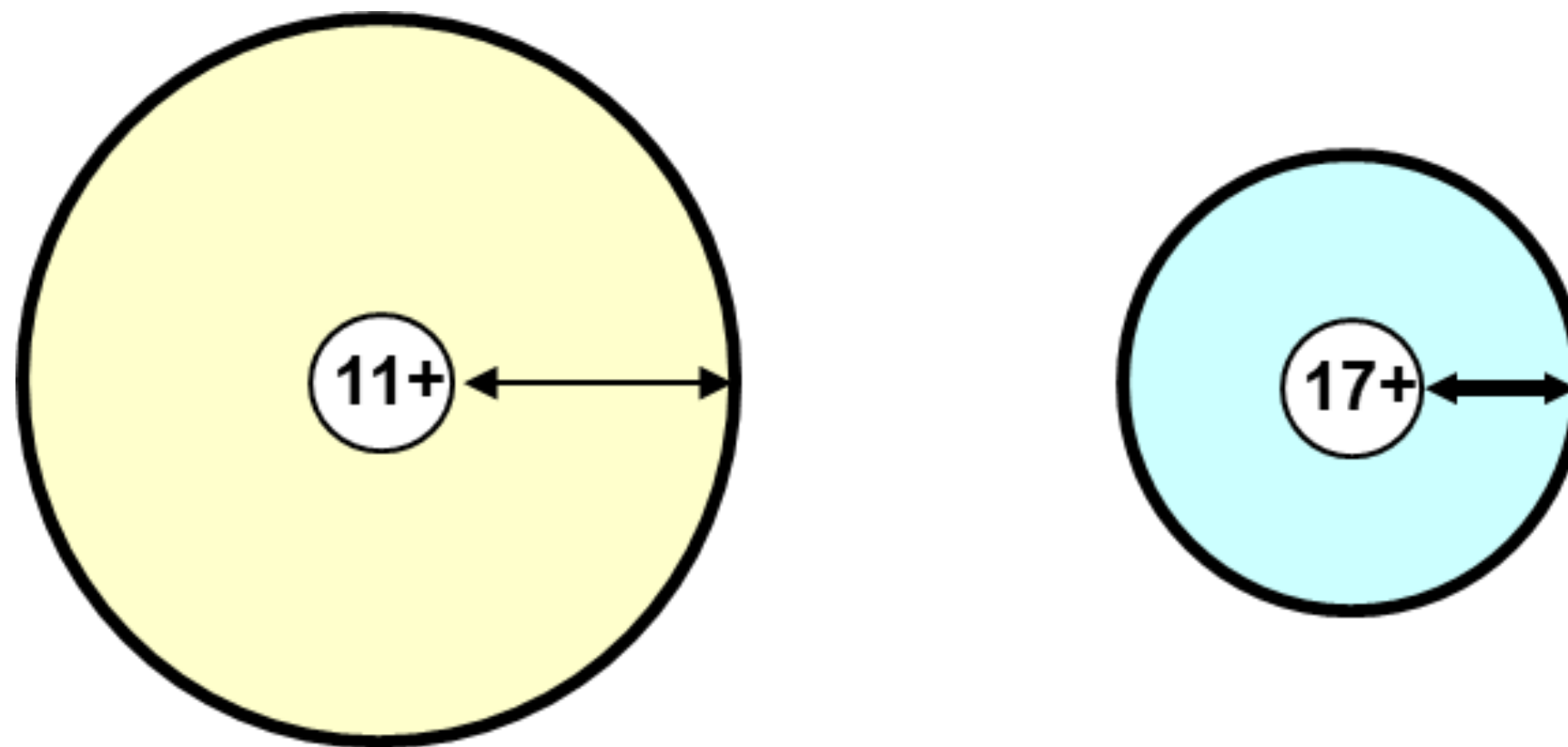
INTRODUCTION

- Many physical and chemical properties are influenced by the outer shell configuration of an atom, including:
 - atomic radius
 - ionic radius
 - ionization energy
 - electron affinity
 - Electronegativity
 - electrical conductivity
 - melting point and boiling point
 - It must be noted though that the first two periods in the periodic table are not typical:
 - Period 1 - contains only two elements
 - Period 2 - elements at the top of each group have small sizes and high ionization energy values
 - As such, Period 3 (Na through Ar) is the most suitable period for studying trends
-

TRENDS – ATOMIC RADIUS

General Trend

- Decreases across a given period (i.e., going left to right across Period 3, from Na to Ar)
- As we move from left to right across the period, the nuclear charge increases by +1 each time (as we add a proton). As the nuclear charge increases, it has a greater attraction for the electrons and pulls them in slightly.



TRENDS – 1st IONIZATION ENERGY

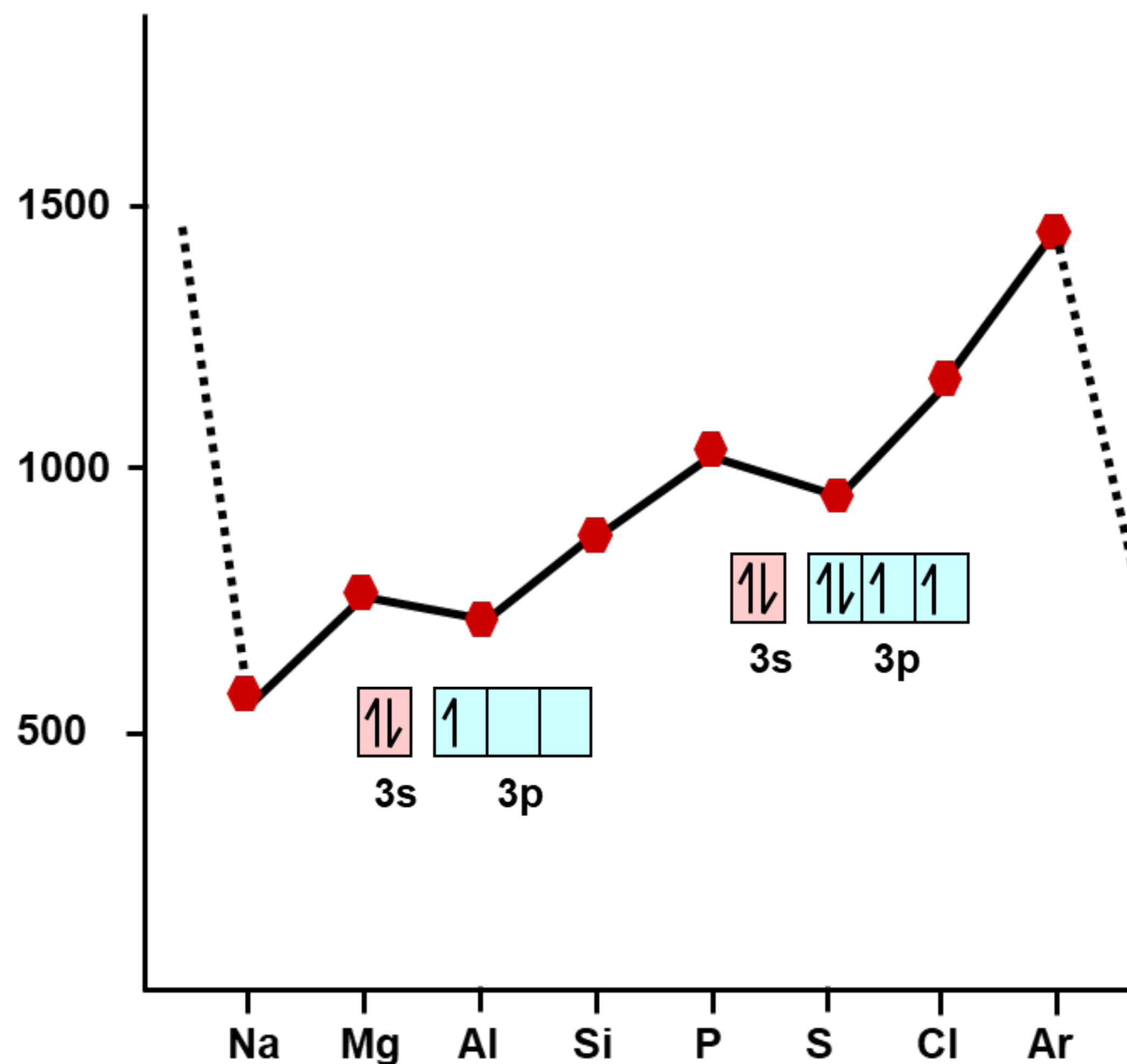
Definition

- A measure of the energy required to remove an outer shell electron from a gaseous atom. Electrons are negatively charged and are attracted to the positively charged nucleus and electrons that are held more strongly will require more energy to overcome the attraction.

General Trend

- **Increases across a given period** (i.e., going left to right across Period 3, from Na to Ar), as electrons are held more strongly and are harder to remove.
 - The nuclear charge increases by one each time, as we move across the Period, each extra electron, however, is going into the same main energy level so is subject to similar shielding and is a similar distance away from the nucleus.
 - It should be noted that the trend is not consistent. Theoretically, the value should increase steadily across the period due to the increased nuclear charge; however, there is a **drop** in the value for aluminum because the extra electron has gone into a 3p orbital. The increased shielding makes the electron easier to remove. There is also a **drop** in the value for sulphur. The extra electron has paired up with one of the electrons already in one of the 3p orbitals. The repulsive force between the electrons means that less energy is required to remove one of them.
-

TRENDS – 1st IONIZATION ENERGY



	NUCLEAR CHARGE	3s	3p
Na	11+	1	
Mg	12+	1↓	
Al	13+	1↓	1
Si	14+	1↓	1 1
P	15+	1↓	1 1 1
S	16+	1↓	1↓ 1 1
Cl	17+	1↓	1↓ 1↓ 1
Ar	18+	1↓	1↓ 1↓ 1↓

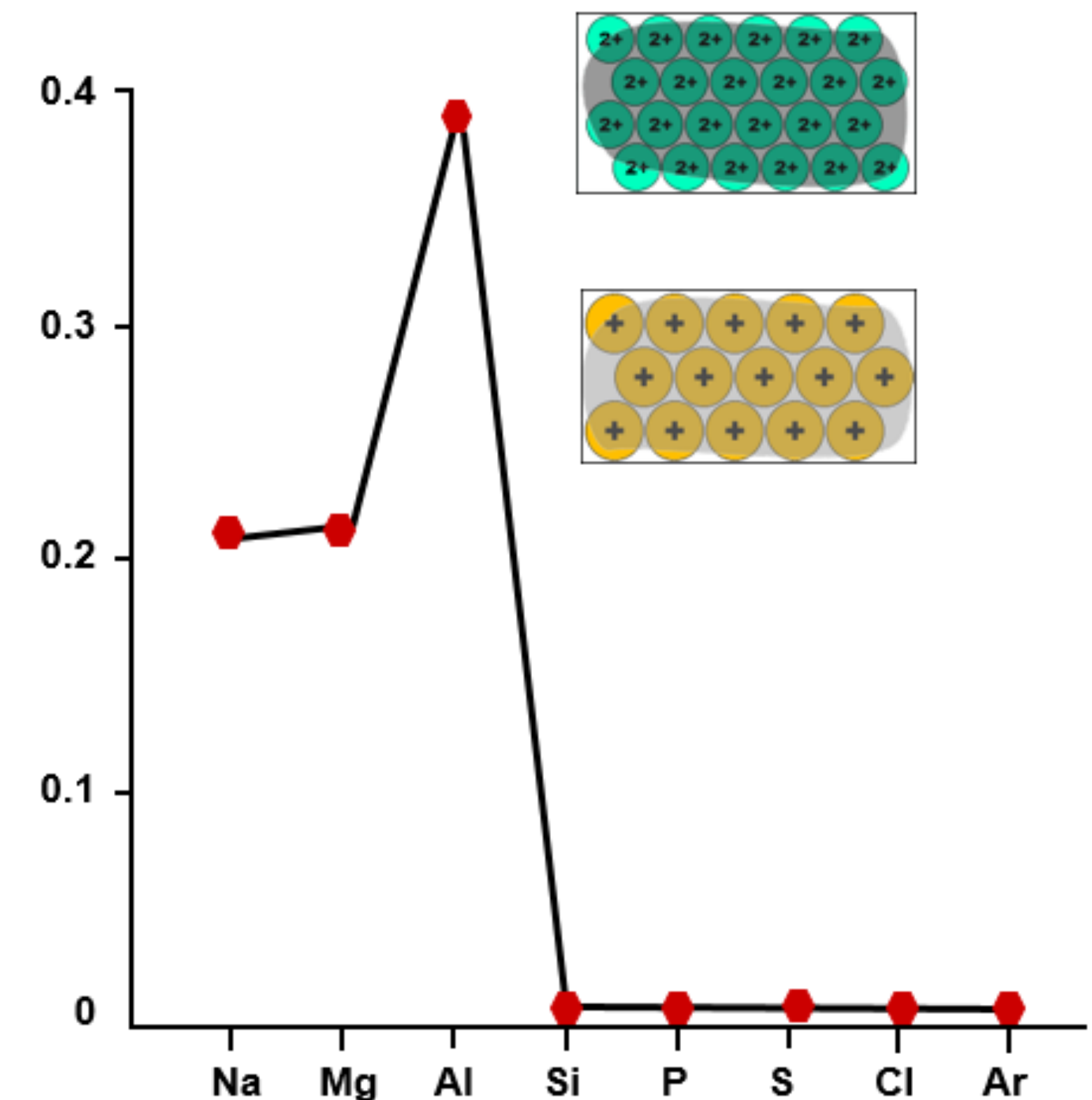
TRENDS – ELECTRICAL CONDUCTIVITY

Definition

Substances conduct electricity when ions or electrons are free to move.

General Trend

- **Decreases across a given period** (i.e., going left to right across Period 3, from Na to Ar):
 - **For Na, Mg, Al:** metallic bonding occurs with delocalized electrons
 - **For Si, P, S, Cl:** covalent bonding occurs, and no electrons are free to move; and
 - **For Ar:** monatomic (noble gas) and electrons are held very tightly.
- **Decreases down a group** where there is any electrical conductivity



UNITS:- Siemens per metre

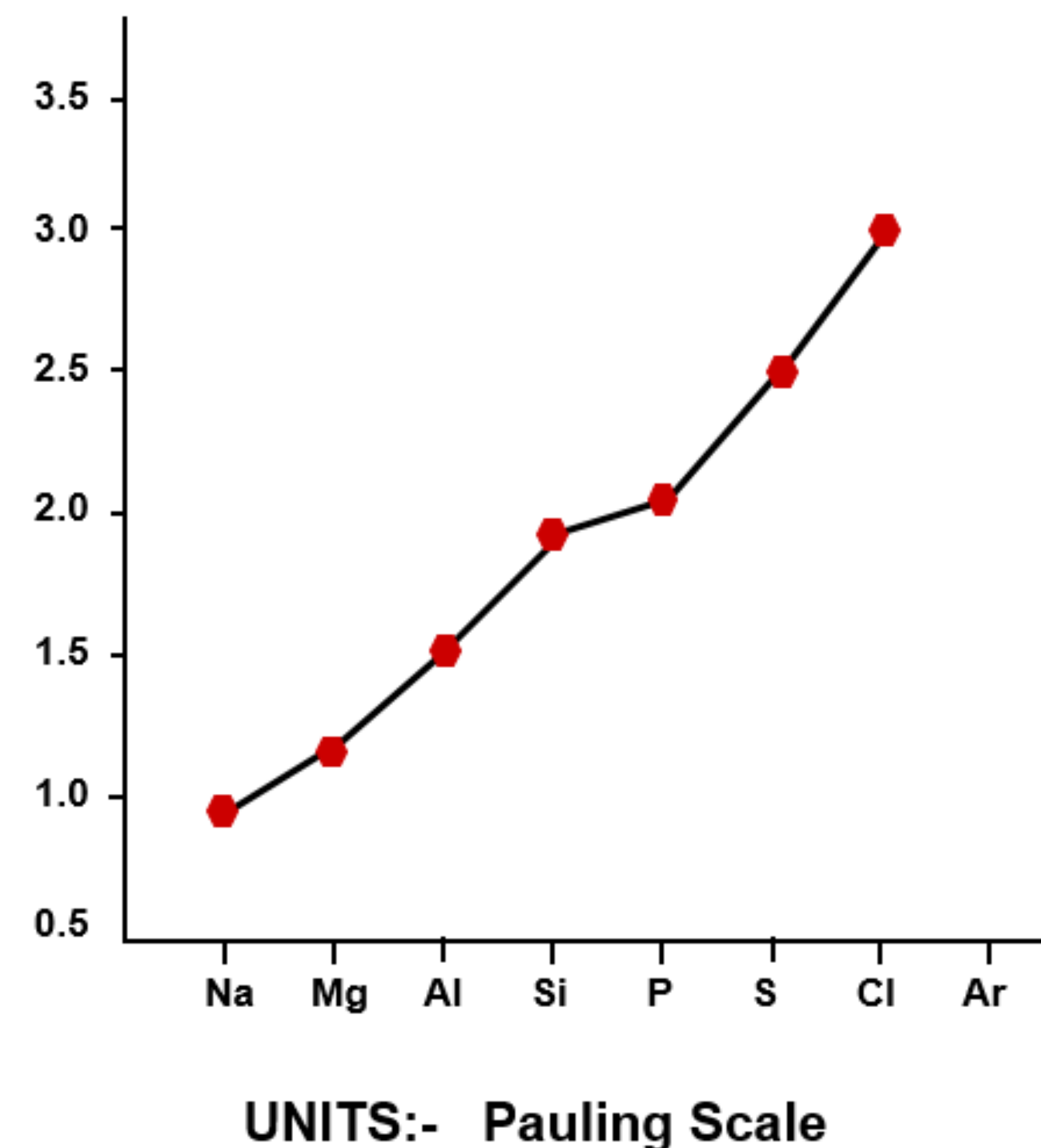
TRENDS - ELECTRONEGATIVITY

Definition

A measure of the attraction an atom has for the pair of electrons in a covalent bond. (Do not confuse with electron affinity.)

General Trends

- **Increases across a period** because the nuclear charge is increasing and therefore so does the attraction for the shared pair of electrons in a covalent bond; and
- **Decreases down a group** because although the nuclear charge is increasing, the effective nuclear charge is less due to shielding of filled inner shells and a greater distance from the nucleus.



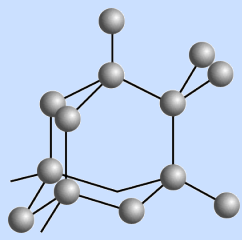
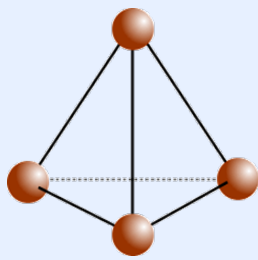
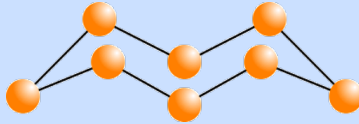
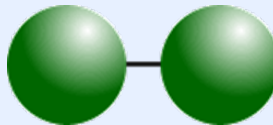
TRENDS – MELTING POINT

Definition

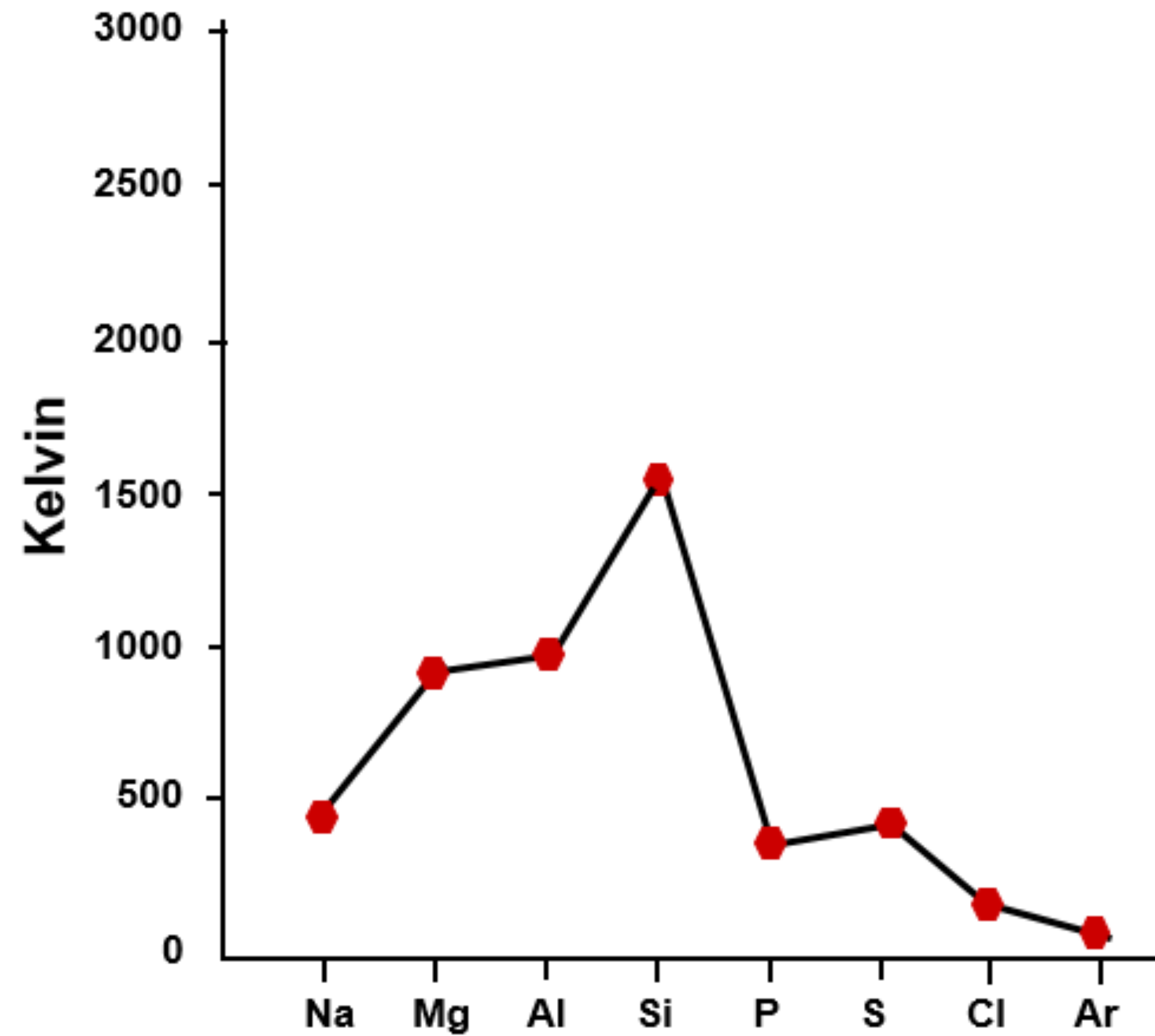
Melting point is a measure of the energy required to separate the particles in a substance from solid to liquid; bond type plays a significant role in the required energy.

General Trend

- Increases moving across a period, and then decreases;
 - For Na, Mg, Al:** the melting point increases due to the increasing strength of metallic bonding.
 - For Si:** large increase in melting point as it has a giant molecular and a lot of energy is required to break the many covalent bonds holding the atoms together
 - For P, S, Cl, Ar:** much lower melting points as they are simple covalent molecules (typically P_4 , S_8 , Cl_2) and their melting point depends on the weak intermolecular Van Der Waals' forces (Note: the larger the molecule, the greater the Van Der Waals' forces)

Element	Structure	Relative Mass	Melting Point (°C)
Si		NA	1414
P_4		124	44
S_8		256	119
Cl_2		71	-101
Ar		40	-189

TRENDS – MELTING POINT



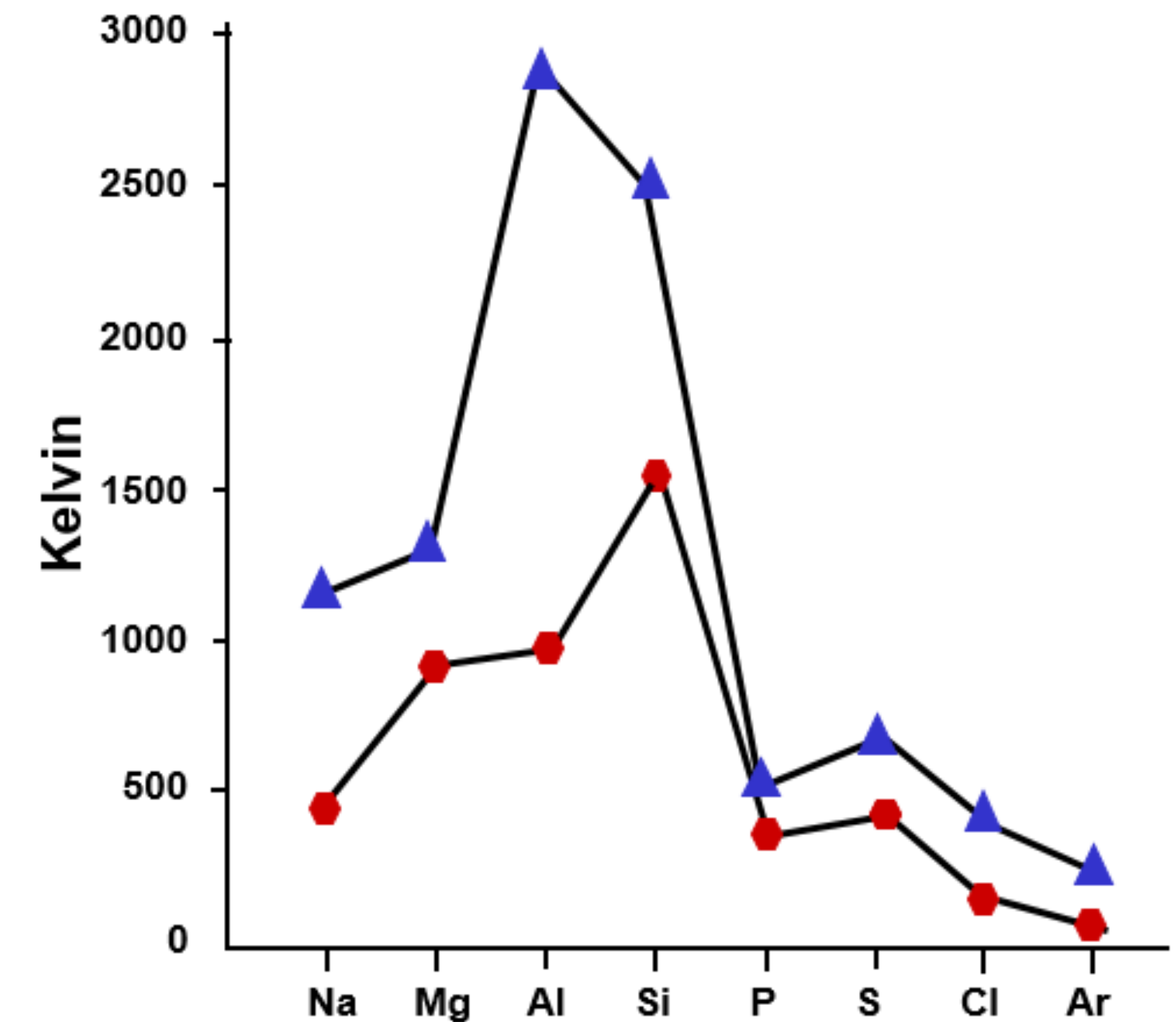
TRENDS – BOILING POINT

Definition

Boiling point is a measure of the energy required to separate the particles in a substance from liquid to gas; bond type plays a significant role in the required energy also.

General Trend

- Increases moving across a period, and then decreases (same as melting point);
- Boiling points tend to be a better measure and show better trends because solids can be affected by the crystal structure as well as the type of bonding.
- As is expected, the boiling points (blue) are higher than the melting points (red).



REREFENCES

1. Chemistry: Atoms First, an OpenStax resource (<https://cnx.org/contents/RTmulxzM@7.152:9H-cauk3@13/Preface>)
2. Chemistry Library, The Living Library (https://chem.libretexts.org/Bookshelves/Introductory_Chemistry)